सी एस आई आर – राष्ट्रीय भौतिक प्रयोगशाला
CSIR-NATIONAL PHYSICAL LABORATORY
(वैज्ञानिक तथा औद्योगिक अनुसंधान परिषद्)
(Council of Scientific and Industrial Research)
(राष्ट्रीय मापिकी संस्थान (एनएमआई), सदस्य बीआईपीएस एवं हस्ताक्षरकर्ता सीआईपीएम - एमआरए)
(National Metrology Institute (NMI), Member BIPM and Signatory CIPM - MRA)
डॉ. के. एस. कृष्णन् मार्ग, नई दिल्ली-110012, भारत
Dr. K. S. Krishnan Marg, New Delhi-110012 INDIA
दूरभाष/Phone : +91-11-4560, 8441, 8610, 9447, फैक्स/Fax : 91-11-4560 8448
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अंशांकन प्रमाण पत्र
CALIBRATION CERTIFICATE
AN ELECTRONIC BALANCE

प्रमाण पत्र संख्या / Certificate No.

N24080687/D1.01/C-082

दिनांक/ Date	अगले अंशांकन हेतु अनुशंसित तिथि Recommended date for next calibration	पृष्ठ / Page	पृष्ठों की संख्या / No. of Pages
01.10.2025	01.10.2028	1	2

1. **Calibrated for** : Fluid Flow Metrology Section
CSIR - National Physical Laboratory
Dr. K. S. Krishnan Marg
New Delhi - 110012

Customer's Ref. No. : CTR Form (Internal) Dated : 27.08.2024

2. **Description and identification of instrument** : An electronic balance with specifications :

Make	:	Mettler Toledo
Model	:	ICS429
Serial No.	:	2738070002
Maximum capacity	:	150 kg
Minimum capacity	:	1.25 kg
Readability (d)	:	5 g
Verification scale interval (e)	:	25 g
Pan design	:	Square

3. **Environmental conditions** : Ambient condition

4. **Standard(s) used with associated Measurement uncertainty** : NPL working standards (Sub-Div.#1.01/M/20kg/00, Sub-Div.#1.01/M/224(2)/00 & Sub-Div.#1.01/M/224(3)/00 of mass with uncertainty better than one-third of the reported uncertainty of measurement.

5. **Metrological Traceability of Standard(s) used** : The working standard(s) used for calibration is(are) traceable to the National Standard which realize the physical units of mass according to the International System of units (SI).

6. **Principle/Methodology of Calibration & Calibration Procedure number** : CSIR-NPL Calibration Procedure No. Sub-Div.#1.01/Doc.3/CP#WI/M-04. The calibration procedure is based on OIML R 76-1 : 2006.

अंशांकनकर्ता:
Calibrated by:
(D.C. SHARMA)

जाँचकर्ता:
Checked by:
(NIDHI SINGH)

प्रभारी वैज्ञानिक:
Scientist-in-charge:
(NIDHI SINGH)

जारीकर्ता:
Issued by: डॉ. श्रीनिवास राव रागम
Dr. Srinivasa Rao Ragam

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01.10.2025	01.10.2028	2	2

7. **Results** :

i. Weighing performance test

The balance was tested for its accuracy, both for progressively increasing and decreasing loads. The indication errors at diffrent loads were found to be as follows:

Denomination (kg)	Error (kg)	MPE (kg)
5	0.010	0.013
10	0.010	0.013
20	0.010	0.025
30	0.010	0.025
40	0.010	0.025
50	0.010	0.025
60	0.010	0.038
70	0.010	0.038

Denomination (kg)	Error (kg)	MPE (kg)
80	0.010	± 0.038
90	0.010	± 0.038
100	0.015	± 0.038
110	0.015	± 0.038
120	0.015	± 0.038
130	0.015	± 0.038
140	0.015	± 0.038
150	0.015	± 0.038

ii. Eccentricity test

Load of 50 kg was successively placed at the center and at ½ distance from the centre over four segments of the pan in clockwise direction, then in counter-clockwise direction and finally at the centre. The maximum changes in indications from the centre to any other point was found to be 0.003 kg against the maximum permissible errors of ± 0.025 kg.

iii. Repeatability test

The balance was tested for repeatability of its indication at 50 kg, 100 kg & 150 kg loads. The differences between the maximum and minimum indications for the above loads were found to be 0.015 kg, 0.020 kg & 0.020 kg respectively against the maximum permissible errors of ± 0.025 kg, ± 0.038 kg & ± 0.038 kg respectively. The standard deviations were found to be 0.005 7 kg, 0.006 5 kg & 0.005 4 kg respectively.

iv. Uncertainty of measurement : ± 0.005 kg for all the loads.

The reported uncertainty is at coverage factor k = 2 which corresponds to a coverage probability of approximately 95 % for a normal distribution.

8. **Date(s) of Calibration** : 06.09.2024

9. **Remarks** : The balance was calibrated in Fluid Flow Measurement Lab.

अंशांकनकर्ता:
Calibrated by:
(D.C. SHARMA)

जाँचकर्ता:
Checked by:
(NIDHI SINGH)

प्रभारी वैज्ञानिक:
Scientist-in-charge:
(NIDHI SINGH)

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